

Luke Whitaker

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Professional Summary

Multilingual researcher and data specialist with expertise in annotation, data quality, and applied linguistics. Skilled at generating and validating structured datasets under detailed guidelines, with strong analytical judgment honed through published academic research. Proficient in Python, experienced with LLM evaluation workflows, and fluent in English, French, and Spanish. Passionate about the interface of Language and technology. Committed to high-standards, detail-oriented work.

Core Competencies

- **Languages:** English (native), French (professional fluency), Spanish (professional fluency)
- **Data Annotation & Quality Assurance:** Guideline-based coding, inter-rater reliability, data validation
- **Programming & Data Tools:** Python (Pandas, NumPy, Scikit-learn), R, JupyterLab, HTML/CSS, JavaScript
- **Research & Analysis:** Experimental design, survey methodology, statistical modeling (regression, ANOVA, mixed effects)
- **AI & LLM Evaluation:** Prompt engineering, LLM output assessment, generative AI integration, policy analysis
- **Process Improvement:** Annotation guidelines development, workflow documentation, tool adoption
- **Cross-functional Collaboration:** Technical team coordination, stakeholder communication, feedback loops

Professional Experience

Data Analyst – AI & Language Research: *The future of language education (2024)*

University of Iowa

- Designed annotation guidelines and coding schemas for a large-scale national survey on AI writing tools, defining structured categories for qualitative and quantitative response data.
- Cleaned and analyzed 100+ responses using **Python (Pandas, NumPy)**, performed **linear regression modeling (OLS, mixed-effects)** in JupyterLab, and visualized key trends with Seaborn and Matplotlib.
- Built and validated a structured dataset with coded variables (e.g., experience level, AI usage frequency), auditing entries for consistency and establishing reusable annotation standards.
- Collaborated with technical team to synthesize findings into actionable reports, translating complex statistical results into clear insights for diverse stakeholders.
- Trained and evaluated an LLM-based classification model on student-written responses, reviewing model outputs against human annotations to assess accuracy and refine classification guidelines.

Instructional Services Specialist

University of Iowa, Dept. of Spanish & Portuguese | Oct. 2022–Present

- Annotated, categorized, and organized a multilingual archive of narrative interviews, applying metadata tagging schemas (language, topic, speaker background) to ensure data quality and rapid retrieval.
- Reviewed and edited multilingual educational content against detailed style and accuracy guidelines, collaborating with subject-matter experts to iteratively improve quality standards.
- Documented processes and created internal guides for tools and platforms, reducing onboarding friction and improving team-wide consistency.

Project Specialist (HR/Payroll SaaS)

Toast Inc. (formerly StratEx) | Apr. 2016–Aug. 2019

- Validated and quality-checked large-scale employee datasets (.csv), identifying and resolving data inconsistencies to ensure integrity across client implementations.
- Created and maintained detailed process documentation and configuration guides, establishing internal quality standards that reduced errors and improved onboarding.
- Trained cross-functional stakeholders on platform use, translating technical requirements into clear, actionable guidance for non-technical teams.

Education

M.A. in Linguistics

University of Iowa, 2021

Thesis: *Heritage French & Interrogatives in Context* (mixed-effects modeling, Qualtrics, Praat)

B.A. in International Politics/Relations

University of Iowa, 2014

Minors: Linguistics, French, Spanish | Exchange: Aix-en-Provence, France

Publications

- [in progress] Zimotti, G., Frances, C., Whitaker, L., Russell, J., & Smith, A.M. (2026). Nature vs nurture: A large-scale, longitudinal study assessing the role of instructional design on the order of L2 Spanish acquisition. *Second Language Research & Practice*.
- [submitted] Zimotti, G., Smith, A.M., Frances, C., Whitaker, L., & Russell, J. (2025) Validating a Can-do self-assessment tool: validity, utility, and student reflections. *Second Language Research & Practice*.
- Zimotti, G., Frances, C., Whitaker, L., et al. (2024). *The future of language education: Teachers' perceptions about the surge of AI writing tools*. Technology in Language Teaching & Learning. <https://doi.org/10.29140/tltl.v6n2.1136>
- Whitaker, L. (2021). *French interrogatives in context*. University of Iowa. <https://doi.org/10.17077/etd.005964>